Qian Ouyang

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Education

Xi'an Jiaotong University 2023 - 2027 (expected)
Undergraduate studies in Artificial Intelligence
Artificial Intelligence Experimental Class

University of Notre Dame August - December 2025
Undergraduate Exchange Student (non-degree)
Semester GPA: 3.917/4.0 | 12 credits
Selected coursework: Computer Vision - graduate level (A); Natural Language Processing (A); Computer Graphics (A).

Research Experience

University of Notre Dame

Research Assistant October 2025 - Present
Prof. Tengfei Luo's group | Department of Aerospace and Mechanical Engineering

Machine learning for polymer simulation

Participated in polymer simulations using UMA / FAIR-Chem and MACE, including preliminary density calculations and comparisons with reference data. Examined GPU-memory constraints and compared runtime, memory use, and allocated GPU time in selected MACE runs.

Representation learning for molecular dynamics

Explored learned representations for coarse-grained polymers and MACE-based atomistic water. Used the simpler water setting to examine how input information and temporal training affect physical prediction, with comparisons against simple baselines. Examined cases where lower training loss did not consistently improve physical predictions.

Summer Research Participant (iSURE) July - September 2026

Course Project

Driving Coach: Lane and Traffic-Object Detection Fall 2025
Computer Vision, University of Notre Dame | Instructor: Prof. Adam Czajka

Adapted an existing YOLOv1-style detector with a pretrained ResNet-18 backbone for lane and traffic-object detection in BDD100K driving scenes. Used an unseen Shenzhen highway video for qualitative evaluation and analyzed errors involving reflections, lane geometry, and traffic signs.

Skills

Programming: Python, C++

Tools: Notre Dame CRC computing environment